Stimulated Hawking Radiation in Polariton BEC: Reservoir-Corrected Predictions from SK-EFT

SK-EFT Hawking Research Program
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The measured Paris-LKB smooth horizon sets a nominal polariton Hawking scale near 85 mK. We compute stimulated gain in an adopted dispersive model and compare cavity decay with the surface gravity. Under saturated greybody transmission, half gain occurs at $\omega/\kappa_{\rm eff} = \ln(3)/(2\pi)$. This gives a single-shot signal-to-noise ratio of five with 100 probe photons per mode under ideal shot-noise scaling. A physical detection prediction requires transmission, calibrated readout and background noise over a usable band.

I. INTRODUCTION

Analog Hawking radiation — the quantum emission of phonon pairs at acoustic horizons in flowing superfluids — has been confirmed in atomic BEC [1, 2] at temperatures $T_H \approx 0.35$ nK. Exciton-polariton condensates offer a $\sim 10^8$-fold temperature advantage, with predicted $T_H \sim 0.1{-}1$ K, due to stronger interactions and sharper spatial gradients achievable in semiconductor microcavities [3, 4].

The 2025 demonstration of all-optically tunable acoustic horizons with tailored curvature in a polariton quantum fluid at LKB Paris [5, 6] — including the first observation of negative-energy Bogoliubov modes in the supersonic (post-horizon) region — establishes the reported horizon and excitation profiles. We evaluate a model for a stimulated measurement at that horizon.

Our analysis incorporates three corrections absent from prior theoretical predictions:

1. **Reservoir correction to c_s**: The excitonic reservoir absorbs 50–75% of the polariton blueshift without contributing to the dynamical speed of sound, reducing c_s by a factor of $\sim 2{-}3$ relative to naive Bogoliubov estimates [7–9].

2. **Stimulated measurement**: CW pump-probe spectroscopy [10] motivates a stimulated gain measurement. The measured gain and its uncertainty depend on transmission, readout transfer and background noise.

3. **Formal checks**: `PolaritonTier1.lean` proves properties of the adopted attenuation factor. The formal checks cover the stated mathematical relationships.

II. RESERVOIR-CORRECTED PARAMETERS

A. Speed of sound

The polariton speed of sound enters the acoustic metric as $c_s = \sqrt{g n_{\rm cond}/m^*}$, where only the condensate density $n_{\rm cond}$ (not the total density including dark excitons) contributes to the dynamical response. Three independent measurements establish:

Group	c_s (μm/ps)	Method
Falque et al. (2025)	0.40	Bogoliubov spectrum
Estrecho et al. (2021)	0.40	Dipole oscillations
Amo et al. (2009)	0.81	Cherenkov cone

TABLE I. Measured polariton speed of sound in GaAs microcavities. The prior theoretical estimate of 1.0 μm/ps [12] was never measured — it traces to a reservoir-free GPE simulation.

We adopt Falque's measured values directly: $c_s = 0.40$ μm/ps with upstream healing length $\xi \approx 3.4$ μm (Falque 2025, §IV.1). The slight offset from $\hbar/(m^* c_s) = 3.75$ μm reflects the finite density of the driven-dissipative condensate; downstream of the horizon $\xi \approx 4.0$ μm.

B. System parameters

For the LKB Paris platform [5], the smooth and steep horizon configurations span a measured range of κ:

Parameter	Smooth baseline	Steep reach	Source
m^*	7.0×10^{-35} kg	–	Falque 2025
c_s	4.0×10^5 m/s (0.40 μm/ps)		Falque 2025 §IV.1
ξ	3.4 μm (upstream)		Falque 2025 §IV.1
κ	7.0×10^{10} s^{-1}	1.1×10^{11} s^{-1}	Falque 2025 Fig. 2 / §IV.2
$D = \xi\kappa/c_s$	0.59	0.94	computed
$-\pi D^2/6$	-0.19	-0.46	dispersive correction
T_H	85 mK	134 mK	$\hbar\kappa/(2\pi k_B)$

TABLE II. Polariton analog horizon parameters for the LKB Paris platform. Both smooth (red / purple traces in Falque Fig. 2) and steep (Falque §IV.2) horizon configurations are reported by the primary source. The smooth-horizon value is used as the baseline throughout this paper; At the steep-horizon input $D = 0.93$, the adopted term $-\pi D^2/6$ is approximately -46%. A physical truncation error remains to be established.

III. STIMULATED HAWKING GAIN SPECTRUM

A. Bogoliubov scattering at the horizon

A coherent probe at frequency ω incident on the analog horizon undergoes anomalous Bogoliubov scattering with amplification

$$G(\omega) = \frac{\Gamma(\omega)}{e^{2\pi\omega/\kappa_{\rm eff}} - 1} \tag{1}$$

where $\Gamma(\omega)$ is the greybody factor. For the model illustration we adopt $\kappa_{\rm eff} = \kappa(1 - c_1 D^2)$ with $c_1 = \pi/6$. At the smooth-horizon baseline $D \approx 0.60$, this gives a model shift of about -19%. At the steep horizon $D \approx 0.93$ the same expression gives about -46%, outside the small-correction regime. A device-specific dispersion law and its error range remain to be established. At unit transmission in the nondispersive limit, $G(\omega) \to 1/(e^{2\pi\omega/\kappa} - 1) = n_{\rm Planck}(\omega, T_H)$.

For an input probe with $n_{\rm in}$ quanta:

$$\langle N_\omega^{\rm out} \rangle = |\alpha_\omega|^2 n_{\rm in} + |\beta_\omega|^2 (n_{\rm in} + 1) \tag{2}$$

where the first term is ordinary scattering, the second is spontaneous + stimulated Hawking emission with $|\beta_\omega|^2 = G(\omega)$.

B. Illustrative probe frequencies

At unit greybody transmission the model gives $G(0.1\kappa_{\rm eff}) \approx 1.14$, $G(0.3\kappa_{\rm eff}) \approx 0.18$, and $G(\kappa_{\rm eff}) \approx 0.002$. Figure 1 uses an adopted upper working edge of $0.3\kappa_{\rm eff}$. A physical error bound and a calibrated lower edge remain to be established.

C. Signal-to-noise ratio

Under ideal shot-noise scaling,

$$\mathrm{SNR}(\omega) = \sqrt{N_{\rm shots} N_{\rm probe}}\, G(\omega). \tag{3}$$

The illustrative case $\omega = 0.1\kappa_{\rm eff}$ with 100 probe photons per mode gives a single-shot model SNR of about 11. A detection prediction requires calibrated transmission, readout transfer and background noise.

IV. PLATFORM COMPARISON

V. DRIVEN-DISSIPATIVE CONSIDERATIONS

The polariton condensate is inherently driven-dissipative. Key implications for Hawking physics:

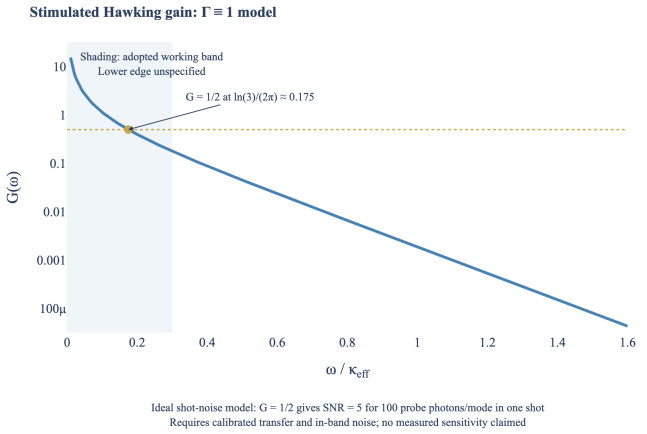


FIG. 1. Unit-greybody model gain against $\omega/\kappa_{\rm eff}$. The dashed line marks half gain at $\ln(3)/(2\pi)$. In the ideal shot-noise model, half gain with 100 probe photons per mode yields a single-shot signal-to-noise ratio of five. The shaded working band illustrates the adopted upper edge.

Platform	T_H	$\tau_{\rm pol}$	$\Gamma_{\rm pol}/\kappa$
Long (projected)	85 mK	100 ps	0.143
Ultralong (projected)	85 mK	300 ps	0.0476
Standard (measured)	85 mK	8 ps	1.79
Steep horizon	134 mK	8 ps	1.14
Steinhauer BEC	0.35 nK	—	—

TABLE III. Registered model parameters. The measured Paris cavity has 8 ps lifetime and $\Gamma_{\rm pol}/\kappa \approx 1.79$. At the same horizon the projected 100 ps and 300 ps cavities give about 0.143 and 0.048, respectively. Only the latter satisfies the adopted weak-loss criterion $\Gamma_{\rm pol}/\kappa < 0.1$. Detection also requires a calibrated noise and readout model. The steep-horizon 134 mK entry is a formal gradient scale for a non-thermal predicted spectrum.

- The acoustic metric description holds in the weak-loss, long-wavelength limit [14]. The ratio $\kappa\tau_{\rm pol}$ compares the cavity lifetime with the inverse surface gravity. A value above one means the cavity lifetime exceeds that characteristic timescale.
- The scale $m^* c_s^2/k_B \approx 0.81$ K is the mean-field energy in temperature units. The stimulated measurement requires an independent background-noise spectrum.
- The cited quasinormal mode studies [11, 15] motivate looking for resonant structure. Using a resonance to choose a probe frequency requires a calibrated response and noise spectrum.

VI. FORMAL VERIFICATION

All numerical predictions in this paper derive from `src/core/formulas.py` and `src/core/constants.py`,

the canonical sources of truth in the SK-EFT pipeline. Specific function traceability:

- $T_H = \hbar\kappa/(2\pi k_B)$: `polariton_hawking_temperature` → `hawking_temp_from_surface_gravity` (`AcousticMetric.lean`)
- $G(\omega) = \Gamma/(\exp(2\pi\omega/\kappa_{\text{eff}}) - 1)$: `stimulated_hawking_gain` (Aristotle: pending)
- SNR $= \sqrt{N_{\text{shots}} \cdot N_{\text{probe}}} \cdot G(\omega)$: `stimulated_hawking_snr`
- $D = \xi\kappa/c_s$: derived in `constants.py` from `POLARITON_PLATFORMS`
- Spatial attenuation ≥ 1: `attenuation_ge_one` (`PolaritonTier1.lean`, manual proof)
- Attenuation monotonicity: `attenuation_mono_Gamma` (`PolaritonTier1.lean`, manual proof)

Parameter provenance: all polariton parameters (c_s, ξ, κ, τ_{cav}, m^*) are registered in `PARAMETER_PROVENANCE` with `llm_verified_date` set and primary source citations. The c_s correction from $1.0 \to 0.40$ μm/ps (adopting Falque's measured value directly, 2026-04-13 Phase 5u Wave 4) was resolved via the Deep Research Reconciliation Protocol (3 independent measurements vs. 1 theoretical projection).

VII. CONCLUSION

The measured smooth-horizon gradient and decay rate fix the nominal temperature and cavity-loss scales used here. The gain curve is conditional on the adopted dispersion and unit transmission. A longer cavity addresses one requirement. The proposed experiment also needs a measured transfer and noise budget over the working band.

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